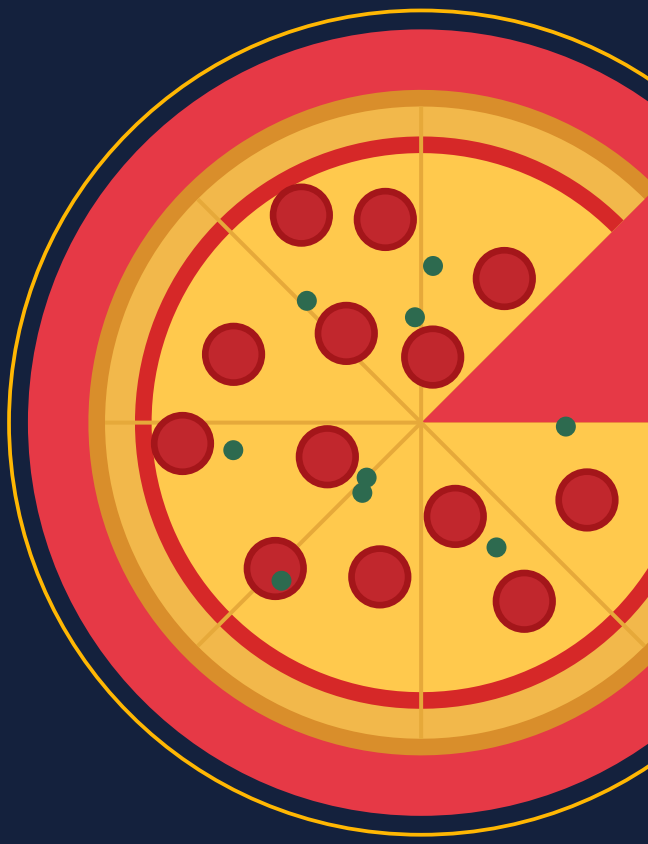


DATA ANALYSIS PROJECT

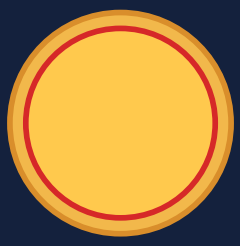
Pizza Hut Sales Analysis

13 business questions solved with MySQL using joins, aggregations, subqueries and window functions.

21,350	$817,860	138
TOTAL ORDERS	TOTAL REVENUE	PIZZAS PER DAY

TOOLS: MySQL Workbench
TABLES: orders, order_details, pizzas, pizza_types





Q1 Total number of orders placed

SQL QUERY

```sql
SELECT
    COUNT(order_id) AS total_orders
FROM orders;
```

RESULT

total_orders
21,350

INSIGHT

21,350 orders were placed across the year.

Q2 Total revenue from pizza sales

SQL QUERY

```sql
SELECT
    ROUND(SUM(order_details.qauntity * pizzas.price), 2) AS total_sales
FROM order_details
JOIN pizzas
    ON pizzas.pizza_id = order_details.pizza_id;
```

RESULT

total_sales
817,860.05

INSIGHT

Total sales came to $817,860.05, about $38.31 per order on average.

Q3 Highest-priced pizza

SQL QUERY

```sql
SELECT
    pizza_types.name,
    pizzas.price
FROM pizza_types
JOIN pizzas
    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

RESULT

name	price
The Greek Pizza	35.95

INSIGHT

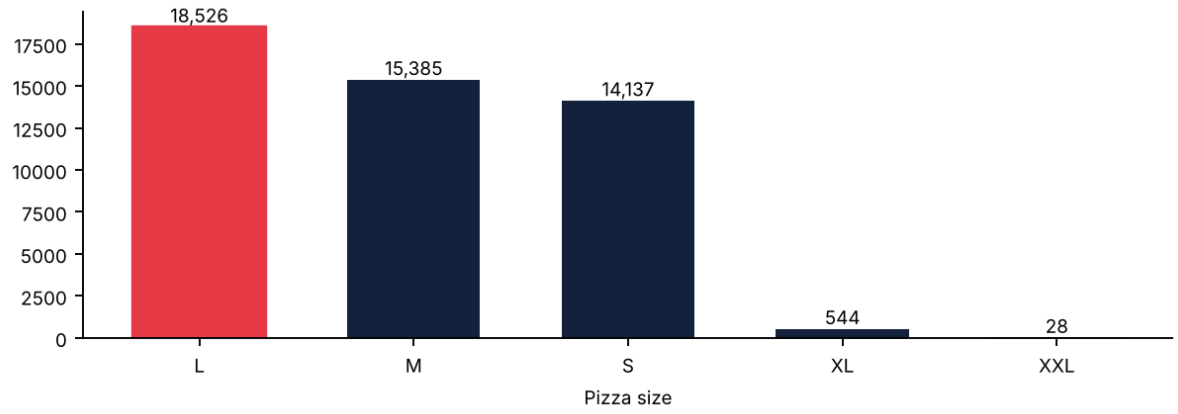
The Greek Pizza is the highest-priced item at $35.95.

Q4 Most common pizza size ordered

SQL QUERY

```sql
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM pizzas
JOIN order_details
    ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

RESULT



size	order_count
L	18,526
M	15,385
S	14,137
XL	544
XXL	28

INSIGHT

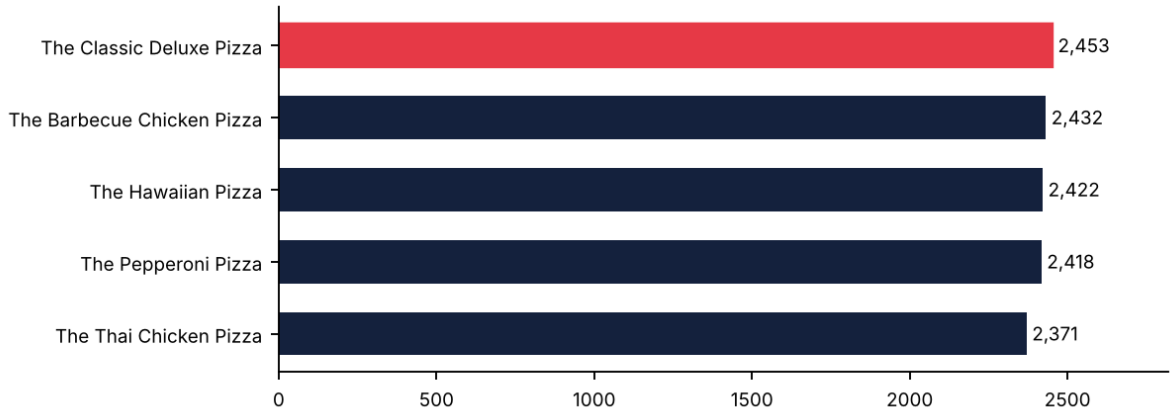
Large (L) is the most ordered size with 18,526 orders (about 38% of all). XXL is very rare at 28 orders.

Q5 Top 5 most ordered pizza types

SQL QUERY

```sql
SELECT
    pizza_types.name,
    SUM(order_details.qauntity) AS quantity
FROM pizza_types
JOIN pizzas
    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
    ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

RESULT



name	quantity
The Classic Deluxe Pizza	2,453
The Barbecue Chicken Pizza	2,432
The Hawaiian Pizza	2,422
The Pepperoni Pizza	2,418
The Thai Chicken Pizza	2,371

INSIGHT

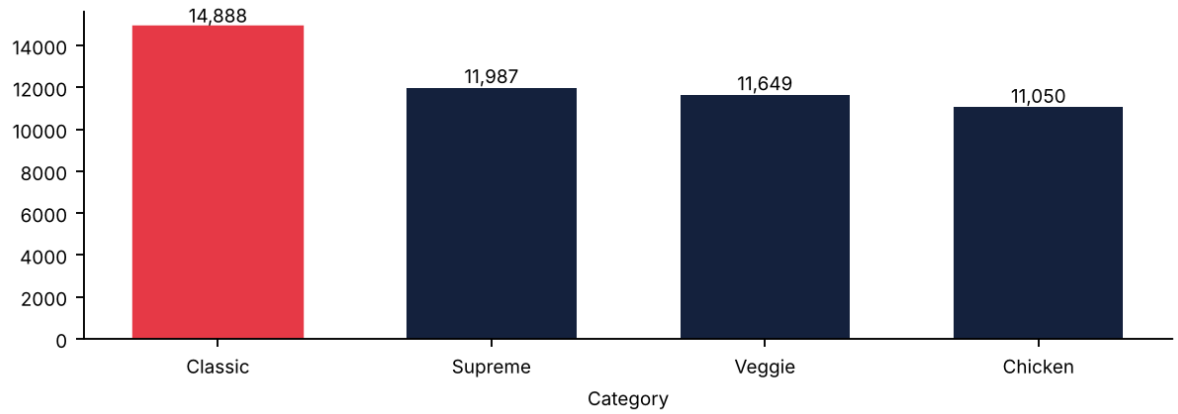
The top 5 are very close (2,371 to 2,453 units). The Classic Deluxe leads, followed by Barbecue Chicken and Hawaiian.

Q6 Total quantity by pizza category

SQL QUERY

```sql
SELECT
    pizza_types.category,
    SUM(order_details.qauntity) AS quantity
FROM pizza_types
JOIN pizzas
    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
    ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

RESULT



category	quantity
Classic	14,888
Supreme	11,987
Veggie	11,649
Chicken	11,050

INSIGHT

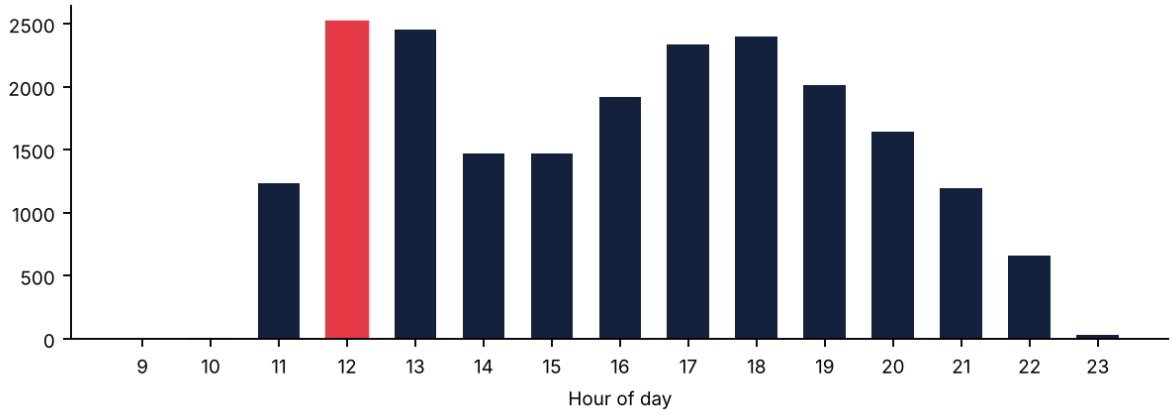
Classic is the best-selling category with 14,888 pizzas (about 30% of the total).

Q7 Orders by hour of the day

SQL QUERY

```sql
SELECT
    HOUR(order_timme) AS Hours,
    COUNT(order_id) AS order_count
FROM orders
GROUP BY HOUR(order_timme);
```

RESULT



Hours	order_count
9	1
10	8
11	1,231
12	2,520
13	2,455
14	1,472
15	1,468
16	1,920

Hours	order_count
17	2,336
18	2,399
19	2,009
20	1,642
21	1,198
22	663
23	28

INSIGHT

Two clear rushes: lunch (12:00-13:00) and dinner (17:00-19:00). Almost no orders before 11 AM or after 10 PM.

Q8 Category-wise distribution of pizzas

SQL QUERY

```sql
SELECT
    category,
    COUNT(name)
FROM pizza_types
GROUP BY category;
```

RESULT

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

INSIGHT

The menu has 32 pizza types: Supreme and Veggie have 9 each, Classic 8, Chicken 6.

Q9 Average pizzas ordered per day

SQL QUERY

```sql
SELECT
    ROUND(AVG(quantity), 0) AS avg_pizzas_orders
FROM (
    SELECT
        orders.order_date,
        SUM(order_details.qauntity) AS quantity
    FROM orders
    JOIN order_details
        ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date
) AS order_quantity;
```

RESULT

avg_pizzas_orders
138

INSIGHT

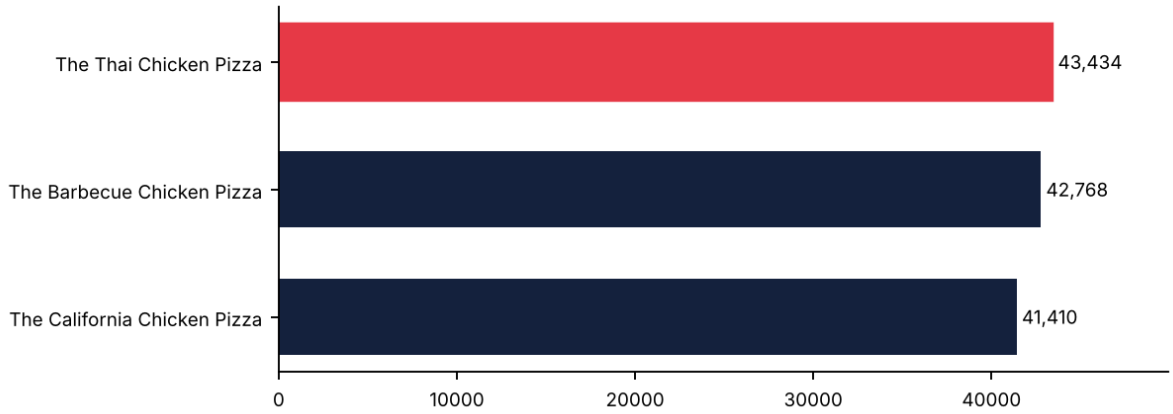
On average, about 138 pizzas are sold per day.

Q10 Top 3 pizza types by revenue

SQL QUERY

```sql
SELECT
    pizza_types.name,
    SUM(order_details.qauntity * pizzas.price) AS revenue
FROM pizza_types
JOIN pizzas
    ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN order_details
    ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

RESULT



name	revenue
The Thai Chicken Pizza	43,434.25
The Barbecue Chicken Pizza	42,768
The California Chicken Pizza	41,409.50

INSIGHT

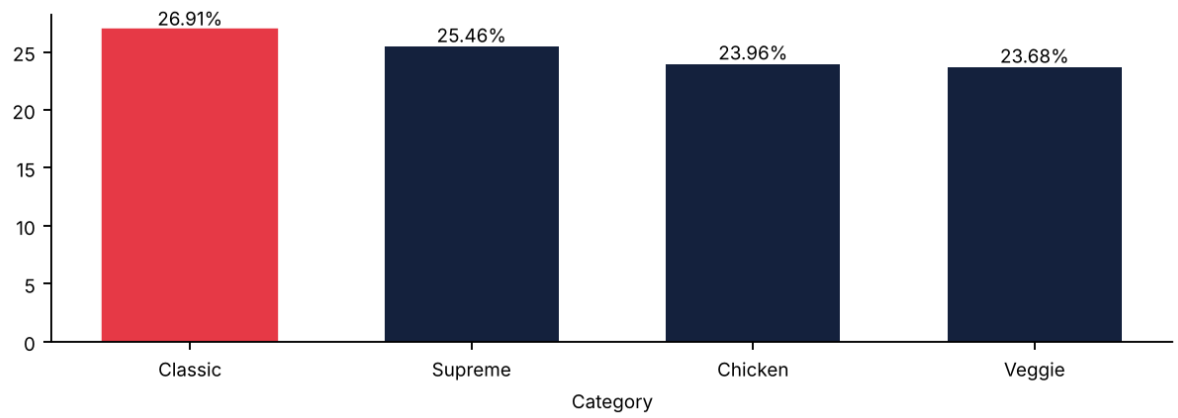
Chicken pizzas take all top 3 revenue spots, led by The Thai Chicken Pizza at $43,434.25.

Q11 Revenue contribution by category (%)

SQL QUERY

```sql
SELECT
    pizza_types.category,
    ROUND(
        SUM(order_details.qauntity * pizzas.price) / (
            SELECT
                ROUND(SUM(order_details.qauntity * pizzas.price), 2)
            FROM order_details
            JOIN pizzas
                ON pizzas.pizza_id = order_details.pizza_id
        ) * 100,
        2
    ) AS revenue
FROM pizza_types
JOIN pizzas
    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
    ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

RESULT



category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

INSIGHT

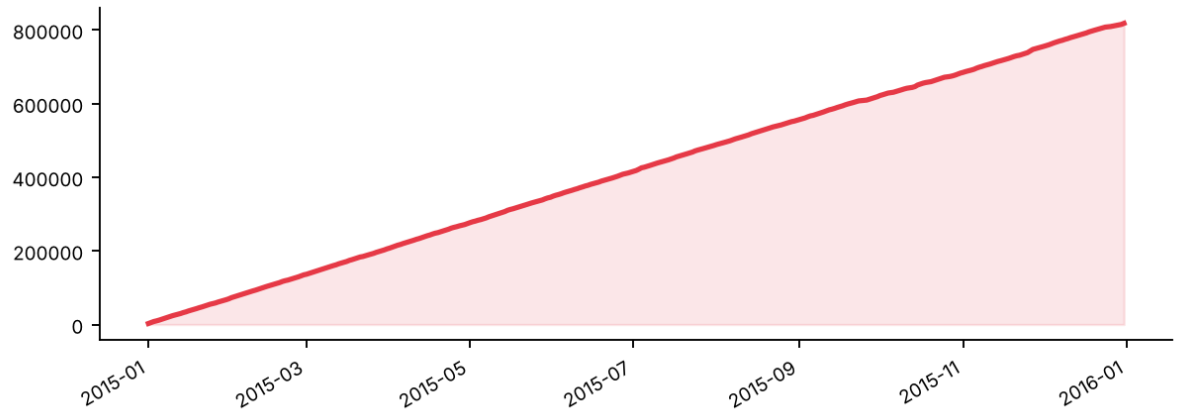
Classic earns 26.91% of revenue, then Supreme 25.46%, Chicken 23.96% and Veggie 23.68%. The four categories are fairly balanced.

Q12 Cumulative revenue over time

SQL QUERY

```sql
SELECT
    order_date,
    SUM(revenue) OVER (ORDER BY order_date) AS cum_revenue
FROM (
    SELECT
        orders.order_date,
        SUM(order_details.qauntity * pizzas.price) AS revenue
    FROM order_details
    JOIN pizzas
        ON order_details.pizza_id = pizzas.pizza_id
    JOIN orders
        ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date
) AS sales;
```

RESULT



order_date	cum_revenue
2015-01-01	2,713.85
2015-01-02	5,445.75
2015-01-03	8,108.15
...	...
2015-12-29	813,606.25
2015-12-30	814,944.05
2015-12-31	817,860.05

358 rows in total, first and last 3 shown.

INSIGHT

Revenue grew steadily through 2015. The final cumulative figure of $817,860.05 matches the total revenue in Q2.

Q13 Top 3 pizza types by revenue in each category

SQL QUERY

```sql
SELECT
    name,
    revenue
FROM (
    SELECT
        category,
        name,
        revenue,
        RANK() OVER (PARTITION BY category ORDER BY revenue DESC) AS rn
    FROM (
        SELECT
            pizza_types.category,
            pizza_types.name,
            SUM(order_details.qauntity * pizzas.price) AS revenue
        FROM pizza_types
        JOIN pizzas
            ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN order_details
            ON order_details.pizza_id = pizzas.pizza_id
        GROUP BY pizza_types.category, pizza_types.name
    ) AS a
) AS b
WHERE rn <= 3;
```

RESULT

category	name	revenue
Chicken	The Thai Chicken Pizza	43,434.25
Chicken	The Barbecue Chicken Pizza	42,768
Chicken	The California Chicken Pizza	41,409.50
Classic	The Classic Deluxe Pizza	38,180.50
Classic	The Hawaiian Pizza	32,273.25
Classic	The Pepperoni Pizza	30,161.75
Supreme	The Spicy Italian Pizza	34,831.25
Supreme	The Italian Supreme Pizza	33,476.75
Supreme	The Sicilian Pizza	30,940.50
Veggie	The Four Cheese Pizza	32,265.70
Veggie	The Mexicana Pizza	26,780.75
Veggie	The Five Cheese Pizza	26,066.50

INSIGHT

Top 3 pizzas by revenue inside each category (category column added for readability).